

CONCEPT BRIEF · DESIGN PHASE

Prediction-Market Quant Vault Marketplace

A two-sided marketplace where quants publish private trading strategies as capital-aligned vaults, and allocators deposit into verified, risk-controlled, real-money track records — trading on Polymarket and Kalshi.

Prepared 27 June 2026 · Configuration **B-on-C** (infrastructure layer + capital allocator)

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1 · The idea

Most people lose money trading prediction markets by hand; a small set of automated, capital-rich, infrastructure-heavy operators win. The opportunity is to package that edge into a product that ordinary allocators can buy into safely — without the operators having to give away their alpha.

What it is

A marketplace with two sides. **Makers** (quants / algo developers) publish trading strategies as **vaults**. **Depositors** allocate capital into the vaults they trust. Strategies run on platform-hosted, sandboxed infrastructure — never on a user's laptop, never holding the keys. Capital is custodied in the protocol. Makers are paid **only when depositors profit**, put their own capital at risk first, and are bounded by automated risk and capacity controls.

What it is deliberately NOT

The naïve version — *"open-source bots, one-click local deploy, decentralized, the moat is the quant network"* — is internally contradictory and fails on four counts. The design below is the corrected form.

Naïve pitch	Why it breaks	This design instead
Open-source strategies	Public + capacity-limited alpha decays to zero; the best quants never list (market for lemons)	Private strategies; verified real-money track record is the public artifact
One-click deploy <i>locally</i>	The only proven-profitable class (latency arb) is an infra game; a laptop is the slow leg; keys unsafe	Hosted , sandboxed execution with a co-located fast path
"Moat = quant network + 1-click"	One-click is copyable; giving away alpha is the opposite of a moat	Moat = verified PnL + connectivity + risk/custody infra
Fee on locked / processed volume	Pays makers to gather & churn even when depositors lose	Profit-only fee + high-water mark + skin-in-the-game

One-line model. Makers earn a profit-only performance fee (HWM, tiered by TVL), post first-loss co-investment + a slashable bond, and live under mechanical slashing and dynamic, realized-data capacity caps. Performance is publicly verified; live positions are disclosed on a lag to protect alpha. Custody is split: on-chain ERC-4626 vaults for Polymarket, a regulated custodial vehicle for Kalshi.

2 · Who it is for

A two-sided market. Each side has a sharp, currently-unmet need.

Supply — Makers (quants)

Independent quants, small algo shops, ex-prop / DeFi-native traders who **have edge but lack capital, infrastructure, and distribution**.

- Want to monetize a strategy **without open-sourcing** it
- Want other people's capital + hosted execution + connectivity
- Want a portable, verifiable **track record**

Today they either run tiny personal size or join a fund and lose ownership.

Demand — Depositors (allocators)

Crypto-native retail & HNW, yield-seekers, and small funds / family offices wanting **uncorrelated, event-driven returns** without building a trading desk.

- The ~90% who **lose** trading prediction markets by hand
- DeFi yield allocators seeking an alternative return stream
- Want vetted, **verified**, risk-controlled, passive exposure

Today their only options are gamble manually or trust an unverified Discord "signal".

Segments & what wins them

Segment	Side	Core need	What the platform gives
Independent quant / algo dev	Maker	Capital + monetization, keep IP	Private strategy, profit-only fee, verified record
Small quant shop / prop team	Maker	Scale beyond own capital	AUM via vaults, hosted low-latency exec
Crypto-native retail / HNW	Depositor	Beat the ~90% loss rate, passively	One-deposit access to vetted strategies
DeFi yield allocator	Depositor	Uncorrelated return stream	On-chain ERC-4626 vault, transparent NAV
Small fund / family office	Depositor	Event-driven alpha, no desk	Regulated wrapper, risk controls, reporting

Beachhead: crypto-native allocators on the Polymarket (on-chain) rail + a handful of invited makers with real track records. The regulated Kalshi rail and institutional depositors come in a later phase.

3 · Probable TAM

Prediction markets are growing fast, but the revenue here is bounded by a hard ceiling: trading profit in these markets is **zero-sum and capacity-constrained**. The pie is the extractable alpha pool, and it grows only as fast as volume, open interest, and "dumb-flow" supply. Honest sizing below.

~\$458B

2025 notional volume, Kalshi + Polymarket (~97.5% of market)

~\$1.08B

Total open interest, late 2025 (6x YoY) — the capacity-relevant number

\$40M+

Documented arbitrage profit extracted, Apr'24–Apr'25 (one strategy class)

Model: revenue = AUM × net alpha × fee take

The platform earns a protocol cut of a profit-only performance fee. The chain is: **extractable alpha pool → AUM that captures it → performance fee → protocol take**.

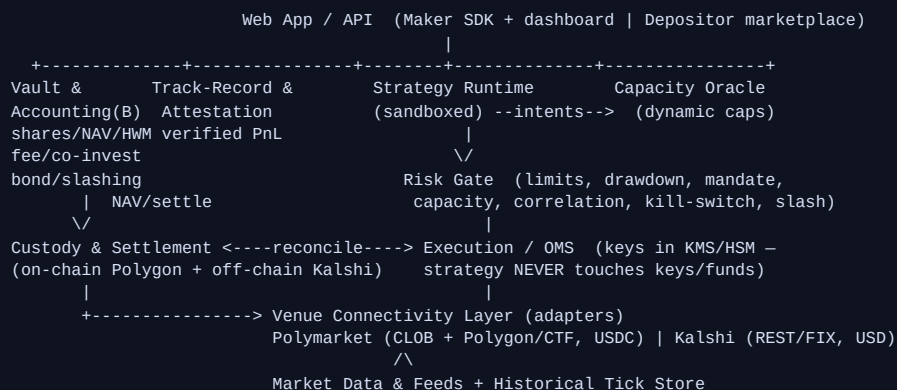
Layer	Definition	Today (est.)	3-yr if market keeps growing
Alpha pool (ceiling)	Total extractable PnL across prediction markets / yr	~\$100–300M	~\$0.5–1.5B
TAM (fee revenue)	~15% blended performance fee on that pool, industry-wide	~\$15–45M/yr	~\$75–225M/yr
SAM	Reachable via Polymarket + Kalshi with the dual-rail structure (~95%+)	~\$14–43M/yr	~\$70–215M/yr
SOM (one platform)	Intermediating \$100–500M AUM; fee pool then protocol take	Fee pool ~\$1.5–11M; protocol rev ~\$0.3–2M	Protocol rev ~\$3–20M

Assumptions: managed/allocator capital ≈ 5–20% of open interest; net alpha to depositors ≈ 10–15%/yr for surviving, capacity-capped strategies; blended performance fee ≈ 15%; protocol take ≈ 15–20% of fees. Ranges, not forecasts.

The honest caveat. This is a **focused, capacity-bound niche**, not a hyperscale market — at today's open interest the whole industry's extractable profit is in the low hundreds of millions. It becomes interesting only on the bet that prediction-market volume and OI keep compounding (monthly volume rose from <\$5B in Sep'25 to ~\$24B by Apr'26). The model scales *with the venues*, and is gated by capacity caps that deliberately cap AUM to protect returns.

4 · Architecture

Configuration **B-on-C**: an infrastructure layer (connectivity, hosted execution, risk, custody, track-record) with a capital-allocator / vault layer on top. Shaped by one rule — **untrusted strategy, trusted platform**.



The security model — the crux

Strategy code is **untrusted**. It runs in an isolated microVM with no key access and no network egress beyond the data feed. It can only emit **intents** (desired orders / target positions). Every intent passes a single **Risk Gate** choke point; only the Execution/OMS — which alone holds keys in a KMS/HSM — signs and routes orders. This neutralizes the "third-party code near my funds" rug risk that sinks the deploy-locally design.

Vault & fee mechanics (capital alignment)

Capital is tranch: **depositor** (senior) · **maker co-investment** (junior / first-loss) · **maker bond** (locked, slashable). Makers earn a **profit-only** fee above a per-share high-water mark, tiered by TVL; the co-invest **ratio** is enforced continuously so alignment can't be diluted as the vault grows. The bond is sized to outweigh the fee's "free option," killing the incentive to take reckless risk.

Custody split & transparency

Polymarket capital sits in **on-chain ERC-4626 vaults**; Kalshi (off-chain, KYC'd, USD) requires a **regulated custodial vehicle**. A unified accounting service consolidates both and reconciles NAV against venue reality (divergence → kill-switch). Verified results are public immediately; **live positions are disclosed on a lag** to prevent copy-trading from decaying the alpha.

Decision	Rationale	Rejected alternative
Hosted, not local	Latency, uptime, key safety, fee enforceability	Local agent (rug risk, slow, no rake)
Intent-based; no keys in strategy	Single validated choke point; no exfiltration / self-dealing	Scoped API keys per strategy
Private + verified track record	Solves alpha decay & market-for-lemons	Open-source marketplace
Profit-only fee + HWM	Ties maker income to depositor outcome	AUM / turnover fee
Co-invest first-loss + slashable bond	Neutralizes the fee's option convexity	Reputation only
Realized-data capacity caps	Stops overcapitalizing capacity-limited edge	Maker-declared capacity

5 · User flows

Maker (quant) flow

- 1 **Onboard & KYC.** Sign up, identity / sanctions checks, accept maker terms.
- 2 **Author & backtest.** Build the strategy in the Maker SDK (Python/TS); run a reproducible, content-addressed backtest against snapshotted historical data.
- 3 **Incubate.** Deploy to an incubating vault with **only the maker's own capital** — no outside AUM yet. Trades run hosted & sandboxed.
- 4 **Earn a verified track record.** Real-money PnL is signed and anchored on-chain (e.g. 60-day minimum) — no backtests count toward eligibility.
- 5 **Unlock outside AUM.** Post the required co-investment (first-loss) + slashable bond; the vault lists publicly with its verified record, capacity cap, and fee tier.
- 6 **Operate & earn.** Collect a profit-only fee above the high-water mark. Stay inside risk limits & capacity caps — breaches trigger mechanical slashing. Graduate to higher TVL tiers as the record lengthens.

Depositor (allocator) flow

- 1 **Browse the marketplace.** Filter vaults by verified performance, Sharpe, max drawdown, capacity headroom, maker co-investment, and fee tier.
- 2 **Diligence a vault.** Inspect the signed real-money track record, lagged position history, risk mandate, and how much skin the maker has in the game.
- 3 **Pass compliance.** KYC + geofencing route the depositor to the correct rail (on-chain Polymarket vs regulated Kalshi vehicle).
- 4 **Deposit.** Funds mint vault shares at the current NAV/share — subject to the capacity cap (excess is rejected or queued).
- 5 **Monitor.** Track NAV, returns, drawdown, and delayed positions live; receive risk-event and slashing notifications.
- 6 **Withdraw.** Redeem shares at NAV after a notice period (positions may be illiquid until event resolution). Performance fee crystallizes **only on profit** above the high-water mark.

Trust loop. Makers can't touch keys or depositor funds; depositors can't see a strategy's live hand in time to copy it; the platform reconciles every vault's NAV to the venue and halts on divergence. Each side gets what it needs without exposing the other.

6 · Feasibility & risks

Verdict: **Conditionally feasible.** The software is buildable by a small team in months; the hard parts are economics, alignment, and regulation — all addressable with the design above, but bounded by three conditions.

Dimension	Status	Note
Demand	Strong	Real & compounding; the least of the problems
Value capture	Solved	Profit-only fee, enforceable via in-protocol custody
Incentive alignment	Solved*	*if co-invest ratio + bond are sized right — the #1 design task
Adverse selection (lemons)	Mitigated	First-loss + slashing + verified record self-select genuine edge
Economics at scale	Bounded	Zero-sum / small-OI ceiling caps total TAM
Regulatory	Heavy	Profit-fee on pooled capital ⇒ CPO/CTA wrapper, front-loaded

The three binding conditions

1. **Co-investment ratio + bond sizing** must provably outweigh the performance fee's option value — model this first; get it wrong and you've built a blow-up machine.
2. **TAM is capped by open interest** (~\$0.2–0.3B/venue today). Caps protect depositors but bound revenue; the thesis rides on continued market growth.
3. **A regulated wrapper** (CPO/CTA + custodial Kalshi rail, geofenced Polymarket rail) is mandatory and front-loaded.

Phased build

Phase	Scope	Proves
0	Infra + Polymarket only + one internal strategy, no outside AUM	Infra + a real track record
1	Vaults + profit-only fee + HWM + co-invest, invited makers, capped AUM	Economics + alignment
2	Slashing + dynamic capacity caps + public track record	Safety + trust
3	Kalshi rail + CPO/CTA wrapper	Multi-venue + compliance
4	Open marketplace	Scale (within the OI ceiling)

Sources for market figures: Pew Research; The Block; CoinDesk; KuCoin; academic prediction-market arbitrage analyses (Apr 2024–Apr 2025); QuantConnect Alpha Streams postmortem. Figures are estimates for planning, not investment advice.